# 1 Exercise 1

Let $S$ be the set {rock, paper, scissors}. Define a binary operation $* : S \times S \to S$ via

$$x * y = \begin{cases} \text{the winner of a rock-paper-scissors game between } x \text{ and } y, & \text{if } x \neq y \\ x, & \text{if } x = y. \end{cases}$$

(a) Prove that $*$ is a commutative operation on $S$.

(b) Prove that $*$ is <u>not</u> an associative operation on $S$.

## 1.1 Solution (a)

*Proof.* Commutativity means that $x * y = y * x$.

There are two cases: when $x = y$ and $x \neq y$. Begin with the former. If $x = y$, $x * y = x$ and $y * x = y$. By the transistive property, since $x * y = x$ and $x = y$, $x * y = y$. Also by the transistive property, since $x * y = y$ and $y * x = y$, $x * y = y * x$. Ergo, if $x = y$, $x * y$ is commutative.

Now the case where $x \neq y$. Without loss of generality, suppose that $x$ has the advantage over $y$ (i.e. $x$ is rock when $y$ is scissors, $x$ is scissors when $y$ is paper, etc.). In this case, $x$ will always beat $y$ in a game of ro-sham-bo, so $x * y = x$. Meanwhile, $y * x$ also results in a game of ro-sham-bo, which $x$ has the advantage in and will win, so $y * x = x$. Since $x * y = x$ and $y * x = x$, by the transistive property, $x * y = y * x$.

Ergo, $*$ is a commutative operation on $S$. OE∆

## 1.2 Solution (b)

*Proof by Counterexample.* For the associative property to hold, it must be the case that for any $r, p$, and $s$ in $S$, $(r * p) * s = r * (p * s)$. Suppose that $r =$ rock, $p =$ paper, and $s =$ scissors. We can perform the internal operations on them. Start with $(r * p) * s$.

$$(r * p) * s = p * s = s = \text{scissors} \tag{1}$$

Then, $r * (p * s)$.

$$r * (p * s) = r * s = r = \text{rock} \tag{2}$$

For associativity to hold, it would require that $s = r$, meaning scissors = rock, which is expressly not true. That acts as a conterexample, so associativity does not hold. OE∆

## 2 Exercise 2

Let $F$ be an ordered field with additive identity 0 and multiplicative identity 1.

(a) Using **only** the field and ordered field axioms, prove that if $a, b \in F$ with $a > 0$ and $0 < b < 1$, then $0 < ab < a$.

(b) Using **only** the field axioms, prove that if $a, b, c, d \in F$, then $((a+b)+c)+d = d+(c+(b+a)) = ((d+c)+b)+a$.

### 2.1 Solution (a)

*Proof.* The multiplicative ordered field axiom states that if for some $a$ and $b$ in $F$, $a \leq b$ and $c \geq 0$, then $ac \leq bc$.

Since $0 < b$ and $0 < a$, $0 \leq b$ and $0 \leq a$. In the same vein, since $b < 1$, $b \leq 1$.

By the multiplicative axiom, $0 \cdot a \leq ab$ and $ab \leq a$.

We proved that $0 \cdot x = 0$ in class, so $0 \cdot a = 0$ and $0 \geq ab$.

We proved in class that if $x \neq 0$ and $y \neq 0$, then $xy \neq 0$. Since $0 < a$, $a \neq 0$. Since $0 < b$, $b \neq 0$. Therefore, $ab \neq 0$. Since $0 \leq ab$ and $0 \neq ab$, $0 < ab$.

We proved in class that if $x \neq 0$ and $xy = xz$, then $y = z$. Its contrapositive is that if $y \neq z$, then $x = 0$ or $xy \neq xz$. Since $b < 1$, $b \neq 1$. This means that either $a = 0$ or $ab \neq a$. Since $a \neq 0$, $ab \neq a$.

$ab \leq a$ and $ab \neq a$, so $ab < a$.

Since $0 < ab$ and $ab < a$, $0 < ab < a$. OEΔ

### 2.2 Solution (b)

*Proof.* Our first half of the proof will consist of a long series of equalities, all of which are justified by the commutative property. The commutative property of addition says that $a + b = b + a$.

$$
\begin{aligned}
((a+b)+c)+d &= ((b+a)+c)+d && (3)\\
&= (c+(b+a))+d && (4)\\
&= d+(c+(b+a)) && (5)
\end{aligned}
$$

Therefore, $((a+b)+c)+d = d+(c+(b+a))$.

The second half of the proof will consist of a long series of equalities, all of which are justified by the associative property. The associative property of addition says that $(a+b)+c = a+(b+c)$, or equivalently that $a+b+c$ is unambiguous.

$$
\begin{aligned}
d+(c+(b+a)) &= (d+c)+(b+a) && (6)\\
&= ((d+c)+b)+a && (7)
\end{aligned}
$$

Therefore, $d+(c+(b+a)) = ((d+c)+b)+a$.

Ergo, $((a+b)+c)+d = d+(c+(b+a)) = ((d+c)+b)+a$. OEΔ

# 3 Exercise 3

Let $F$ be a field with order $<$. Recall from class that $(F, <)$ is called an *ordered field* if the following axioms are satisfied:

(i) For all $a, b, c \in F$ with $a \leq b$, we have $a + c \leq b + c$.

(ii) For all $a, b, c \in F$ with $a \leq b$ and $0 \leq c$, we have $ac \leq bc$.

(a) Prove that axiom (ii) is *equivalent* to the axiom

(ii)' For all $a, b \in F$ with $a > 0$ and $b > 0$, we have $ab > 0$

That is, prove that (ii) is true if and only if (ii)' is true. This means that whether we use (ii) or (ii)' in our definition of an ordered field is irrelevant.

(b) Define a relation $\prec$ on $\mathbb{Q}$ by declaring $a \prec b$ if and only if $b < a$ in the usual order on $\mathbb{Q}$. Prove that $\prec$ is an order on $\mathbb{Q}$ and that axiom (i) is true for $\prec$, but that axiom (ii) fails. Thus $(\mathbb{Q}, \prec)$ is an ordered set but not an ordered field.

*Note:* The LaTeX commands for $\prec$, $\preceq$, $\succ$, and $\succeq$ are `\prec`, `\preceq`, `\succ`, and `\succeq`, respectively.

## 3.1 Solution (a)

*Proof.* First the forward direction, meaning (ii) implies (ii)'.

Take (ii), and suppose we set $a = 0$, rename $b$ to $a$, and rename $c$ to $b$. This may be confusing, but what we end up with is the following:

(ii) For all $a, b \in F$ with $0 \leq a$ and $0 \leq b$, we have $0b \leq ab$.

We proved in class that $0x = 0$. Therefore, $0b = 0$ and we can replace $0b$ with just 0.

(ii) For all $a, b \in F$ with $0 \leq a$ and $0 \leq b$, we have $0 \leq ab$.

Since this applies for all $a$ and $b$ in $F$, including those that are not equal to 0, we can set the rule of only considering cases where $a \neq 0$ and $b \neq 0$. Furthermore, we proved in class that if $x \neq 0$ and $y \neq 0$, then $xy \neq 0$. Since $a \neq 0$ and $b \neq 0$, $ab \neq 0$. Thereby, we can replace the $\leq$ signs with just $<$.

(ii) For all $a, b \in F$ with $a > 0$ and $b > 0$, we have $ab > 0$.

This gets us to (ii)'.

Now for the backwards direction.

Since $a > 0$ and $b > 0$, without loss of generality, $0 < a \leq b$. Focus on the latter two terms, $a \leq b$. Part (i) tells us we can add $-a$, the additive inverse of $a$, to all sides and end up with $a - a \leq b - a$. By the additive inverse axiom, we know that $a - a = 0$. Therefore, $0 \leq b - a$.' Let there also be some $0 \leq c$. From here, we bifurcate our proof between (1) $0 < b - a$ and $0 < c$, (2) $0 < b - a$ and $0 = c$, (3) $0 = b - a$ and $0 < c$, and (4) $0 = b - a$ and $0 = c$ cases.

(1) First, $0 < b - a$ and $0 < c$. By the axiom (ii)', since $0 < b - a$ and $0 < c$, $0 < c(b - a)$. By the distributive property, $c(b - a) = bc - ac$, so $0 < bc - ac$. The additive inverse axiom tells us that there exists some $-(-ac)$ such that $ac + (-(-ac)) = 0$. Part (i) tells us we can add $-(-ac)$ to all sides, so $-(-ac) < bc - ac - (-ac)$. In class, we proved that $-(-x) = x$, so $-(-ac) = ac$,

which we can replace in the equation to get $ac < bc - ac + ac$. The commutative property lets us say that $ac < bc + ac - ac$. By the additive inverse axiom, $ac - ac = 0$. Therefore, $ac < bc + 0$. By the additive identity, $bc + 0 = bc$, so $ac < bc$, and by extension $ac \leq bc$.

(2) Second, $0 < b - a$ and $0 = c$. If we multiply $(b - a)$ by $c$, it is the equivalent to multiplying $(b - a)$ by 0. This means that $c(b - a) = 0(b - a)$. In class, we proved that for all $x$, $0x = 0$. This means that $c(b - a) = 0(b - a) = 0$. The distributive property says that $c(b - a) = bc - ac$. Putting this together, we find $bc - ac = 0$. We can add $-(-ac)$, rewritten as $ac$, to both sides to find $bc - ac + ac = ac$. By the commutative property, $bc + ac - ac = ac$. By the additive identity axion, $bc = ac$, and by extension $ac \leq bc$.

(3) Third, $0 = b - a$ and $0 < c$. If we multiply $(b - a)$ by $c$, it is the equivalent to multiplying $c$ by 0. This means that $c(b - a) = c \cdot 0$. By the commutative property fo multiplication, $c(b - a) = (b - a)c$ and $c \cdot 0 = 0c$, so $(b - a)c = 0c$. In class, we proved that for all $x$, $0x = 0$. This means that $(b - a)c = 0c = 0$. The distributive property says that $(b - a)c = bc - ac$. Putting this together, we find $bc - ac = 0$. We can add $-(-ac)$, rewritten as $ac$, to both sides to find $bc - ac + ac = ac$. By the commutative property, $bc + ac - ac = ac$. By the additive identity axion, $bc = ac$, and by extension $ac \leq bc$.

(4) Last, $0 = b - a$ and $0 = c$. We can add $-(-a)$, which we proved in class is equal to $a$, to $0 = b - a$ to get $a = b - a + a$. The commutative property tells us that $a = b + a - a$. The additive inverse axiom says that $a = b + 0$. The additive identity axiom says that $b = b + 0$, so $a = b$. Multiply both sides by $c$ to get $ac = bc$, and by extension $ac \leq bc$.

Ergo, this proves that if $a \leq b$ and $c \geq 0$, $ac \leq bc$. OE$\Delta$

### 3.2 Solution (b)

*Proof.* The textbook says that $\mathbb{Q}$ is an ordered set if $r > s$ is defined to mean that $r - s$ is a positive rational number. I will start by using this to prove that $\mathbb{Q}$ is an ordered field.

Every number $q$ in $\mathbb{Q}$ can be written in terms of one integer $a$ and one natural number $b$ in the form $q = \frac{a}{b}$. Let there be three rational numbers $\frac{a}{b}$, $\frac{c}{d}$, and $\frac{e}{f}$ such that $\frac{a}{b} < \frac{c}{d}$. Cross-multiply these fractions, bearing in mind that $b$ and $d$ are both positive non-zero integers, to get $ad < bc$. Mutiply both sides by $f^2$ to get $adf^2 < bcf^2$. Add the integer $bdef$ to both sides to get $adf^2 + bdef < bcf^2 + bdef$. Apply the distributive property by $df$ on the left side and $bf$ on the right side to get $df(af + be) < bf(cf + de)$. Divide both sides by $bf$ and $df$ to get $\frac{af+be}{bf} < \frac{cf+de}{df}$. Separate both fractions into two and cancel out terms to get $\frac{a}{b} + \frac{e}{f} < \frac{c}{d} + \frac{e}{f}$. This means that the rational numbers satisfy the first ordered field axiom.

Let there be three rational numbers $\frac{a}{b}$, $\frac{c}{d}$, and $\frac{e}{f}$ such that $\frac{a}{b} \leq \frac{c}{d}$ and $\frac{e}{f} \geq 0$. Since $\frac{e}{f} \geq 0$, we can multiply both sides by $f$ to get $\frac{ef}{f} \geq 0f$. Since we established in class that $0f = 0$, $\frac{ef}{f} \geq 0$. By the multiplicative inverse, $\frac{f}{f} = 1$, so $\frac{ef}{f} = e \cdot 1 \geq 0$. By the multiplicative identity, $e \cdot 1 = e \geq 0$. Therefore, $e \geq 0$ and $f > 0$, and both are integers. Multiply both sides of $e \geq 0$ by $f$ to get $ef \geq 0f = 0$ by what was established in class. Therefore, $ef \geq 0$. Take our initial inequality of equations $\frac{a}{b} \leq \frac{c}{d}$ and cross-multiply to get $ad \leq bc$. Multiply both sides by $ef$ to get $adef \leq bcef$. Divide both sides by $bf$ and $df$ to get $\frac{ae}{bf} \leq \frac{ce}{df}$. These fractions can be separated to find $\frac{a}{b} \cdot \frac{e}{f} \leq \frac{c}{d} \cdot \frac{e}{f}$. This means that the rational numbers satisfy the first ordered field axiom, and by extension that the rational number are an ordered field.

We start by proving that $\prec$ is an order on $\mathbb{Q}$. We know that either $a < b$, $a = b$, or $a > b$. Based on that and that $a > b$ means $a \prec b$, we find that either $a \succ b$, $a = b$, or $a \prec b$.[1] Next, suppose that $a \succ b$ and $b \succ c$. This means that $a < b$ and $b < c$. We know from the first ordered

[1]Could expand on this entire paragraph.

field axiom that if $a < b$ and $b < c$, then $a < c$, so in this case $a < c$. Because of that, we can say that $a \succ c$. In total, if $a \succ b$ and $b \succ c$, then $a \succ c$. This all means that $\prec$ is an order on $\mathbb{Q}$.

Next, we prove that for all $a, b$, and $c$ in $\mathbb{Q}$ with $a \preceq b$, we have $a + c \preceq b + c$. That is to say, for all $a, b$, and $c$ in $\mathbb{Q}$ with $a \geq b$, we have $a + c \geq b + c$. Take the first field axiom, applied to the rational numbers.

(i) For all $a, b, c \in \mathbb{Q}$ with $a \leq b$, we have $a + c \leq b + c$.

Swap $a$ and $b$.

(i) For all $a, b, c \in \mathbb{Q}$ with $b \leq a$, we have $b + c \leq a + c$.

Swap every $\leq$ with $\succeq$, the two being equivalent since $a \prec b$ if and only if $a > b$ and $a = b$ being possible either way.

(i) For all $a, b, c \in \mathbb{Q}$ with $b \succeq a$, we have $b + c \succeq a + c$.

Ergo, axiom (i) is true for $\prec$.

Axiom (ii) can be proven impossible by counterexample. First, we restate the axiom for $<$.

(ii) For all $a, b, c \in F$ with $a \leq b$ and $0 \leq c$, we have $ac \leq bc$.

Replace $<$ with $\prec$ and $\leq$ with $\preceq$.

(ii) For all $a, b, c \in F$ with $a \preceq b$ and $0 \preceq c$, we have $ac \preceq bc$.

Based on the definition of $\prec$, we can replace $a \preceq b$ with $a \geq b$, $0 \preceq c$ with $0 \geq c$, and $ac \preceq bc$ with $ac \geq bc$.

(ii) For all $a, b, c \in F$ with $a \geq b$ and $0 \geq c$, we have $ac \geq bc$.

Suppose that $a = 2$, $b = 1$, and $c = -1$. Since $2 \geq 1$, $a \geq b$. Since $0 \geq -1$, $0 \geq c$. $ac$ is equal to $-2$ and $bc$ is equal to $-1$. Since $-2 < -1$, $ac < bc$, which means that $ac \not\geq bc$, which contradicts the second field axiom. As such, axiom (ii) fails.

Ergo, $(\mathbb{Q}, \prec)$ is an ordered set but not an ordered field. OE$\Delta$

# 4 Exercise 4

Let $S \subset \mathbb{R}$ be nonempty and bounded from above. Let $a = \sup S$.

(a) Show that, for any real $\epsilon > 0$, there is $x \in S$ with $x > a - \epsilon$.

(b) Suppose $a \notin S$. Show that, for any real $\epsilon > 0$, there are infinitely many elements of $S$ in the open interval $(a - \epsilon, a)$.

## 4.1 Solution (a)

*Proof by Contradiction.* Suppose for contradiction that, for any real $\epsilon > 0$, there does not exist $x \in S$ with $x > a - \epsilon$. This means that, for all $x \in S$, $x \leq a - \epsilon$. This means that $a - \epsilon$ is an upper bound of $S$.

We know that $\epsilon > 0$. Add $a$ to both sides of $S$, which through the first ordered field axiom and the fact that $\mathbb{R}$ is an ordered field means that $a + \epsilon > a$. By the additive inverse property of fields, there exists some $-\epsilon$ such that $\epsilon + (-\epsilon) = 0$. Add $-\epsilon$ to both sides of $a + \epsilon > a$ to get $a + \epsilon - \epsilon > a - \epsilon$. By the additive inverse property of fields, $a + \epsilon - \epsilon = a > a - \epsilon$.

Therefore, there exists an upper bound of $S$ that is less than $a$ in the form of $a - \epsilon$. Therefore, $a$ is not the least upper bound of $S$. This contradicts that $a$ is the least upper bound of $S$, so our initial assumption must be false. Ergo, for any real $\epsilon > 0$, there is $x \in S$ with $x > a - \epsilon$. OE$\Delta$

## 4.2 Solution (b)

*Proof.* An infinite number of elements in the interval means that the elements can be mapped onto the natural numbers or a set with a larger cardinality.

We can define a set $Z$ recursively. Suppose that the first element of $Z$, denoted $\zeta_0$, is equal to $a - \epsilon$. As established in part (a), there exists some $x$ in $S$ with $x > a - \epsilon$. We can create a new set $X$ and call the $x$ in $S$ with $x > a - \epsilon$, $x_1$. Since $a$ is the least upper bound of $S$, it is an upper bound of $S$, meaning that because $x_0$ is in $S$, $x_1 < a$.

Now, suppose that we have element $n$ (in the natural numbers) of $x$, denoted $x_n$, which is an element of $S$. Since $x_n$ is an element of $S$ and $a$ is an upper bound of $S$, $x_n < a$. Add the additive inverse of $x_n$, $-x_n$, to both sides, which due to the first ordered field axiom becomes $x_n - x_n < a - x_n$. By the additive inverse definition, $x_n - x_n = 0$, so $0 < a - x_n$. Since $\mathbb{R}$ is a field and fields are closed under addition, $a - x_n$ is a real number. We can denote $a - x_n$ as $\zeta_n$. Since $\zeta_n$ is real and greater than 0, it is possible that some new $\epsilon = \zeta_n$. Now take $a - \zeta_n$. As established in part (a), there exists some $x$ in $S$ with $x > a - \zeta_n$. We can call the $x$ in $S$ with $x > a - \zeta_n$ that we just discovered existed $x_{n+1}$.

This pairs up our subset of the set of elements of $S$ in the open interval $(a - \epsilon, a)$ with the natural numbers. This means that the cardinality of the natural numbers is less than or equal to the cardinality of the set of elements of $S$ in the open interval $(a - \epsilon, a)$. The cardinality of the natural numbers is countably infinite. Ergo, the cardinality of the set of elements of $S$ in the open interval $(a - \epsilon, a)$ is infinite. OE$\Delta$

# 5 Exercise 5

For an integer $m \geq 2$, we define $\mathbb{Z}/m\mathbb{Z}$ to be the set $\{0, 1, \ldots, m-1\}$, together with the operations of addition and multiplication modulo $m$. That is, to add two elements of $m$, we add them as we normally would and then take the remainder upon division by $m$. Multiplication is defined similarly. Thus, in $\mathbb{Z}/5\mathbb{Z}$ for example, we have $2 - 4 = 3$ and $4 \cdot 4 = 1$.

You may assume that all of the axioms for addition, multiplication, and distributivity from fields also hold in $\mathbb{Z}/m\mathbb{Z}$ (with 0 and 1 as the identities), *except* for the axiom

($*$) every nonzero element has a multiplicative inverse

which is the subject of this problem.

(a) Show that if $m$ is a composite number, then axiom ($*$) fails.

(b) Show that if $m$ is a prime number, then axiom ($*$) holds. That is, $\mathbb{Z}/p\mathbb{Z}$ is a field with only finitely many elements!

*Hint for part (b):* You may use *Bézout's identity* from number theory: if $a, b \in \mathbb{Z}$ are coprime then there are $x, y \in \mathbb{Z}$ so that $ax + by = 1$. Then use that for a prime $p$, the integers $1, 2, \ldots, p-1$ are all coprime to $p$. You may also use the *Euclidean division theorem*: namely if $b$ is a positive integer, then any $a \in \mathbb{Z}$ may be written uniquely as $a = bq + r$, where $0 \leq r < b$.

## 5.1 Solution (a)

*Proof.* Having a multiplicative inverse in a set means that for any number $a$ in that set there exists some number $a^{-1}$ also in that set for which $a \cdot a^{-1} = 1$. If $m$ is a composite number, that means that there exists some two other integers $j$ and $k$ greater than or equal to 2 such that $m = j \cdot k$.

Without loss of generality, focus on $j$. We know that $j \geq 2$. Since $2 > 1$, by the transistive property (if $j > 2$, otherwise just substitution), $j > 1$. If we add this to, for any $n \geq 0$, $njk = njk$ (which comes from the reflexive property), we get $njk + j > njk + 1$. By the distributive property, $j(nk + 1) > njk + 1$. We also know that $1 > 0$, which can be added to $njk = njk$ to get $njk + 1 > njk + 0 = njk$. Putting these together, we get $j(nk + 1) > njk + 1 > njk$.

This tells us that $njk + 1$ will always be between $njk$ and $njk + j$, which are consecutive multiples of $j$, but will not be equal to either of them so long as $j \geq 2$. Taking the modulus of this past statement of base $jk$, we find that 1 will always be between 0 and $j$, which are consecutive multiples of $j$, but will not be equal to either of them so long as $j \geq 2$. Ergo, in $\mathbb{Z}/m\mathbb{Z}$ where $m$ is composite, the axiom ($*$) will fail. OE$\Delta$

## 5.2 Solution (b)

*Proof.* For the multiplicative identity to hold in set $\mathbb{Z}/p\mathbb{Z}$ for prime $p$, for all integers $b$ with $0 < b < p$, there must exist some integers $x$ and $y$ with $0 < y < p$ such that, in the set of all integers, $by = px + 1$. Before we completely prove this, we prove that any integer multiplied by 10 ends in a zero.

For any integer $S$, $k$ digits long, we can write it as $S = \sum_{i=0}^{k} d_i \times 10^i$, where every $d_i$ is in the set $\{0, 1, 2, \ldots, 9\}$. Multiply the entire number by 10, and we find that $10S = 10 \sum_{i=0}^{k} d_i \times 10^i = \sum_{i=0}^{k} d_i \times 10^{i+1}$. The first value in our new sum is $d_0 \times 10^1$. Since $i$ only increases from there, the exponent of 10 only increases and never reaches the ones digit. Therefore, we can rewrite our sum as $10S = \sum_{i=0}^{k} d_i \times 10^{i+1} + 0 \times 10^0$. Ergo, the new number ends in zero.

Recall Bézout's identity, which says that if integers $a$ and $b$ are coprime, then there exist integers $x$ and $y$ such that $ax + by = 1$. In our set $\mathbb{Z}/p\mathbb{Z}$, $p$ is prime, so it is coprime with every number in the set $\{1, 2, \ldots, p-1\}$. Set $a$ to be equal to $p$, and suppose $b$ is in the set $\{1, 2, \ldots, p-1\}$. This means that there exist integers $x$ and $y$ such that $px + by = 1$. Without loss of generality, if we replace $x$ with $-x$ and use the commutative property, we get $by - px = 1$. Adding $px$ (the additive inverse of $-px$) to both sides, we get $by - px + px = 1 + px$. The commutative property tells us that $by + px - px = px + 1$. The definition of the additive inverse tells us that $by + 0 = px + 1$. The additive identity tells us that $by = px + 1$. In the system $\mathbb{Z}/p\mathbb{Z}$, $p \equiv 0$. By extension, this means that for any integer $n$, $np \equiv 0$. Add 1 to both sides to find that for any integer $n$, $np + 1 \equiv 1$. Suppose that $n = x$. This means that $xp + 1 \equiv 1$. By the commutative property, $px + 1 \equiv 1$. In the system $\mathbb{Z}/p\mathbb{Z}$, since $by = px + 1$, $by \equiv px + 1 \mod p$. Transistively, this means that $by \equiv 1$. If we can ensure that this would hold in a case where $0 < y < p$, then we can prove the axiom holds.

Since there is the requirement of $b < p$, we can multiply both sides by $y$ to find that $by < py$. We want $y < p$. We can multiply both sides of the inequality by $b$ to find that we want $by < bp$. If $y < p$ already, we're in good company and the rest of this proof is moot. If not, we can multiply both sides of $y < p$ by $p$ (which is a positive integer) to find that we want $py < p^2$. Since $by < py$ and we want it to be that $py < p^2$, transistively it is true that $by < py < p^2$. By the Euclidean division theorem, because both $by$ and $bp$ are positive integers, $by$ can be written uniquely as $by = bpq + r$. By the additive inverse, $by - bpq = r$. By the distributive property, $b(y - pq) = r$, so there is an integer multiple of $p$ that is equal to $r$ and furthermore less than $bp$. Furthermore, since $pqy$ is an integer multiple of $p$ and $by$, and we previously established that it would not change that adding or subtracting a multiple of $p$ from an integer would not change its congruency modulo $p$, so we can safely say that $b(y - pq) \equiv 1 \mod p$. That means that if we set $y' = y - pq$, then $by' < bp$, by extension $y' < p$, and $by' \equiv 1 \mod p$. Therefore, in the set $\mathbb{Z}/p\mathbb{Z}$, $by' = 1$. Ergo, axiom $(*)$ holds. OE$\Delta$

## 6 Exercise 6

Let $F$ be a finite field with identities 0 and 1.

(a) Show that there is an integer $n \geq 2$ so that $\underbrace{1+1+\cdots+1}_{n \text{ times}} = 0$. (*Hint*: by the pigeonhole principle, at least two different sums of the form $1+1+\cdots+1$ must be equal)

(b) By using the fact that $\underbrace{1+1+\cdots+1}_{ab \text{ times}} = (\underbrace{1+1+\cdots+1}_{a \text{ times}})(\underbrace{1+1+\cdots+1}_{b \text{ times}})$ for $a, b \geq 2$, show that the *smallest* possible integer $n$ from part (a) is prime.

(c) Using part (a), show that there are no finite ordered fields.

### 6.1 Solution (a)

*Proof.* The key term here is finite. This means that the cardinality of $F$ is some finite number $k$. $F$ being a field, a field axiom says that it is closed under addition. We know that 1, being the multiplicative identity, is contained in $F$. The additive identity 0 is also contained in $F$.

We can prove by induction that any sum $\underbrace{1+\cdots+1}_{n \text{ times}}$ is in $F$. The base case is where $n = 1$ and the sum is just 1, which we know is in $F$. The inductive step coems first from $\underbrace{1+\cdots+1}_{n \text{ times}}$, which we assume is in $F$. Add 1 (in $F$) to this and we get $\underbrace{1+\cdots+1}_{n \text{ times}}+1$, which is in $F$ because $F$ is a field, which is closed under addition. $\underbrace{1+\cdots+1}_{n \text{ times}}+1$ can be rewritten as $\underbrace{1+\cdots+1}_{n+1 \text{ times}}$. Ergo, $\underbrace{1+\cdots+1}_{n \text{ times}}$ being in $F$ implies $\underbrace{1+\cdots+1}_{n+1 \text{ times}}$ being in $F$.

$F$ is finite, and therfore by the Pigeonhole Principle, there must be two sums of the forms $\underbrace{1+\cdots+1}_{m \text{ times}}$ and $\underbrace{1+\cdots+1}_{n \text{ times}}$ such that $m \neq n$ where $\underbrace{1+\cdots+1}_{m \text{ times}} = \underbrace{1+\cdots+1}_{n \text{ times}}$. Without loss of generality, suppose that $m > n$. This is possible because $m$ and $n$ are integers, which are ordered. By the additive inverse axiom, there exists some $-\left(\underbrace{1+\cdots+1}_{n \text{ times}}\right)$ such that $\underbrace{1+\cdots+1}_{n \text{ times}} - \left(\underbrace{1+\cdots+1}_{n \text{ times}}\right) = 0$. We can add this inverse to the earlier inequality to find that $\underbrace{1+\cdots+1}_{m \text{ times}} - \left(\underbrace{1+\cdots+1}_{n \text{ times}}\right) = \underbrace{1+\cdots+1}_{n \text{ times}} - \left(\underbrace{1+\cdots+1}_{n \text{ times}}\right)$. By the earlier additive inverse, $\underbrace{1+\cdots+1}_{m \text{ times}} - \left(\underbrace{1+\cdots+1}_{n \text{ times}}\right) = 0$. We can distribute the negatives in the second term to get $\underbrace{1+\cdots+1}_{m \text{ times}} + \left(\underbrace{-1-\cdots-1}_{n \text{ times}}\right) = 0$. Since $m > n$, we can pair up every number from the latter term with one from the former term, ending up with $\underbrace{(1-1)+\cdots+(1-1)}_{n \text{ times}} + \underbrace{1+\cdots+1}_{m-n \text{ times}} = 0$. By the additive inverse, $1 - 1 = 0$, so $\underbrace{0+\cdots+0}_{n \text{ times}} + \underbrace{1+\cdots+1}_{m-n \text{ times}} = 0$. The additive identity tells us that $a + 0 = 0 + a = a$, so we can say that $\underbrace{1+\cdots+1}_{m-n \text{ times}} = 0$.

Since $m > n$, $m - n > 0$. Integers being closed under addition, $m - n$ would have to be an integer. Therefore, since $m - n > 0$, the minimum possible integer it could be is 1, so $m - n \geq 1$. Now, suppose that $m - n = 1$. That would mean that $0 = \underbrace{1+\cdots+1}_{m-n \text{ times}} = \underbrace{1+\cdots+1}_{1 \text{ time}} = 1$. This would mean that $1 = 0$, which is impossible in a field, so so $m - n \neq 1$. The next smallest integer that $m - n$ could be is 2, so $m - n \geq 2$. Ergo, for integer $p = m - n \geq 2$, $\underbrace{1+\cdots+1}_{p \text{ times}} = 0$. OE$\Delta$

### 6.2 Solution (b)

*Proof by Contradiction.* Suppose for contradiction that $n$ is composite. Suppose also that $n$ is the smallest integer such that $\underbrace{1+\cdots+1}_{n \text{ times}} = 0$. There exists some two integers $a$ and $b$ such that $n = ab$ and (without loss of generality between $a$ and $b$) $1 < a \leq b < n$. If $n$ is prime, then one of $a$ or $b$ is equal to 1, and the other is equal to $n$.

Take the case of $\underbrace{1+\cdots+1}_{n \text{ times}}$. By the fact we are given and our earlier statement that $n = ab$, we can say that $\underbrace{1+\cdots+1}_{n \text{ times}} = \underbrace{1+\cdots+1}_{ab \text{ times}} = \underbrace{(1+\cdots+1)}_{a \text{ times}}\underbrace{(1+\cdots+1)}_{b \text{ times}}$. We can say that $\underbrace{1+\cdots+1}_{a \text{ times}} = \alpha$ and $\underbrace{1+\cdots+1}_{b \text{ times}} = \beta$. Therefore, $\underbrace{(1+\cdots+1)}_{a \text{ times}}\underbrace{(1+\cdots+1)}_{b \text{ times}} = \alpha\beta$.

In class, we proved that if $x \neq 0$ and $y \neq 0$, then $xy \neq 0$. The contrapositive of this says that if $xy = 0$, then $x = 0$ or $y = 0$. Since $\alpha\beta = 0$, then $\alpha = 0$ or $\beta = 0$. This means that either $\underbrace{1+\cdots+1}_{a \text{ times}} = 0$ or $\underbrace{1+\cdots+1}_{b \text{ times}} = 0$.

According to the textbook, the rationals are an ordered field (p.7). That means that if $x > 0$ and $y < z$, then $xy < xz$ (p.8). We know that $b \geq 2$, meaning $b > 1$. We also know that $a \geq 2$ meaning $a > 0$. Therefore, $a < ab$. We can use the same logic with swapped variables to show that $b < ab$. We earlier established that either $\underbrace{1+\cdots+1}_{a \text{ times}} = 0$ or $\underbrace{1+\cdots+1}_{b \text{ times}} = 0$. Since both $a$ and $b$ are less than $ab$ and $n = ab$, that means that there is some number $c$ (which is either $a$ or $b$) less than $n$ such that $\underbrace{1+\cdots+1}_{c \text{ times}} = 0$. This contradicts that $n$ is the *smallest* integer for which the property holds. Therefore, $n$ cannot be composite. Ergo, $n$ is prime.

The reason that this does not work when $n$ is prime is that the only integer divisors of $n$ are 1 and $n$. It cannot be that $\underbrace{1+\cdots+1}_{1 \text{ time}} = 0$ because $\underbrace{1+\cdots+1}_{1 \text{ time}} = 1$, which would mean that $1 = 0$, cannot be true. If $\underbrace{1+\cdots+1}_{n \text{ times}} = 0$, that is just circular logic, so it's already true and gives us no new information. OE$\Delta$

### 6.3 Solution (c)

*Proof by Contradiction.* Suppose for contradiction that $F$ is an ordered field. Since an ordered field is an ordered set which satisfies a handful of axioms, that means $F$ is an ordered set.

In part (a), we proved that in a finite field, there exists some number $n \geq 2$ such that

$\underbrace{1+\cdots+1}_{n \text{ times}} = 0$. Part (b) established that there is a smallest integer $n$ where that is the case. Associatively, $\underbrace{1+\cdots+1}_{n \text{ times}}$ is equal to $\underbrace{1+\cdots+1}_{n-1 \text{ times}}+1$.

One of the axioms of the ordered fields is that if $x, y$, and $z$ are in $F$ and $y < z$, then $x+y < x+z$. We can prove by induction that $0 < \underbrace{1+\cdots+1}_{n \text{ times}}$. We know that $0 < 1$, which acts as our base case. For the inductive step, we assume that $0 < \underbrace{1+\cdots+1}_{k \text{ times}}$ and add $\underbrace{1+\cdots+1}_{n \text{ times}}$ to both sides of $0 < 1$. We end up with $0+\underbrace{1+\cdots+1}_{k \text{ times}} < 1+\underbrace{1+\cdots+1}_{k \text{ times}}$. By the commutative property and the additive identity being 0, $\underbrace{1+\cdots+1}_{k \text{ times}} < \underbrace{1+\cdots+1}_{k \text{ times}}+1$. We previously established that $\underbrace{1+\cdots+1}_{k \text{ times}}+1 = \underbrace{1+\cdots+1}_{k+1 \text{ times}}$. Transistively, this means $0 < \underbrace{1+\cdots+1}_{k+1 \text{ times}}$. Therefore, for all natural numbers $k$ (which integers $k \geq 2$ is a subset of), $0 < \underbrace{1+\cdots+1}_{k \text{ times}}$.

Suppose now that for some $n \geq 2$, $0 = \underbrace{1+\cdots+1}_{n \text{ times}}$, which we established as possible in part (a). According to our inductive proof, for that $n \geq 2$, $0 < \underbrace{1+\cdots+1}_{n \text{ times}}$. Both of these being true contradicts the first axiom of the ordered sets, which says that one and only one of the statements in the set $\{x < y, x = y, y < x\}$ is true. Therefore, our initial assumption is false. Ergo, $F$ is not an ordered field and no finite ordered field exists. OEΔ